

Physical properties

When the need arises to identify or value a gem, it is sent to a professional gemologist who is certified by a professional gemological body. The gemologist will examine the gemstone for various physical and optical properties in order to make an identification and evaluate its quality.

An essential quality of a gemstone is durability. A gemstone's physical properties determine how durable it is, and how susceptible it is to wear, breakage, and deterioration, as well as the quality of its colour. Note that gems with good cleavage (see right) can be very hard but still be susceptible to cracking.



Hardness

One of the principal determinants of durability is the gem's hardness, or the relative ease or difficulty with which it can be scratched. Measured on the Mohs scale, hardness does not equate to strength, as very hard minerals can also be quite brittle. Gemstones below 5 on the scale are too soft for wear and even stones of 6 or 7 will scratch and abrade.